

REVIEW ARTICLE

COVID-19 changed our world: A systematic review

Kuljeet Singh¹, Agar Chander Pushap², Gurinderjeet Kaur³, Shikha Bharti⁴, Arjumand Jhon⁴, Srishty Sudershan⁵, Feroze Ahmed Dar⁶, Bashir Ahmad Sheikh⁶, Mehraj Bashir⁶, Showkat Ahmad Najar⁶, and Amrit Sudershan^{7*} 

¹Department of Computer Science, School of Sciences, CHRIST (Deemed to be University), Delhi-NCR, Uttar Pradesh, India

²Department of Education, Dakshina Bharat Hindi Prachar Sabha, Chennai, Tamil Nadu, India

³Department of Computer Science and Engineering, Thapar Institute of Engineering and Technology, Patiala, India

⁴Department of Zoology, Lovely Professional University, Phagwara, Punjab, India

⁵Department of Zoology, Central University of Jammu, Jammu and Kashmir, India

⁶Department of Zoology, Sri Pratap College Srinagar, Cluster University of Srinagar, Kashmir, Jammu & Kashmir, India

⁷Department of Human Genetics, Sri Pratap College Srinagar, Cluster University of Srinagar, Kashmir, Jammu & Kashmir, India

Abstract

The coronavirus disease 2019 (COVID-19) pandemic profoundly reshaped political, religious, economic, social, and financial institutions across the globe. More than 80 countries imposed border closures, shut down business activities, and advised citizens to self-quarantine. The pandemic also caused schools to close indefinitely, affecting approximately 1.5 billion children worldwide. This study sought to integrate the diverse effects of COVID-19 on varied human interests and offer a comprehensive overview of the effects of the pandemic. A systematic literature review encompassing the period between February 2019 and December 2022 was conducted using Web of Science, PubMed, and Google Scholar. The search included keywords related to the impact of COVID-19 on health, education, economics, and the environment, and articles were selected based on pre-defined inclusion criteria. The literature review was limited by its specific time frame, potential publication bias, and language restrictions. Nevertheless, the study findings highlight critical areas for prospective policy interventions and recommend that future research endeavors should address the diverse effects of the pandemic across varied sectors. This literature review revealed that COVID-19 exerted far-reaching consequences beyond health, impacting mental and emotional well-being, education, financial and economic stability, social dynamics, business operations, travel and tourism, and the environment. The pandemic introduced a profoundly challenging period of human history, revealing itself as more than just a health crisis: it signified a significant threat to multiple facets of global life. Despite the successful rollout of the severe acute respiratory syndrome coronavirus-2 vaccine, an ongoing public debate persists regarding whether COVID-19 will reemerge or eventually wane.

Keywords: Coronavirus disease 2019; Genomic susceptibility; Economy; Education; Environment; Mental health and impact

Academic editor:

Mihajlo Jakovljevic M.D. Ph.D. MAE

*Corresponding author:

Amrit Sudershan
(amritsudershan0@gmail.com)

Citation: Singh, K., Pushap, A.C., Kaur, G., Bharti, S., Jhon, A., Sudershan, S. *et al.* (2024). COVID-19 changed our world: A systematic review. *Global Health Econ Sustain*.
<https://doi.org/10.36922/ghes.3992>

Received: June 20, 2024

Accepted: September 14, 2024

Published Online: December 9, 2024

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1. Introduction

The global balance was disrupted when severe acute respiratory syndrome coronavirus-2 (SARS-CoV-2) emerged in Wuhan, China, in December 2019 and resulted in the coronavirus disease 2019 (COVID-19) pandemic (Lai *et al.*, 2020). The virus spread quickly, prompting the World Health Organization (WHO) to declare it a public health emergency of international concern on January 30, 2020 (Chan *et al.*, 2020; Li *et al.*, 2020). Coronaviruses were already known but the SARS-CoV-2 marked the third major outbreak of respiratory illnesses caused by these viruses (Zhong *et al.*, 2003). SARS-CoV-2 is considered zoonotic, which means it can spread from animals to humans. It has been linked to varied animals and birds including camels, pigs, turkeys, mice, dogs, and bats (Geller *et al.*, 2012; Anthony *et al.*, 2017). This virus spreads from animals to humans through airborne particles and disperses from person to person primarily through droplets, infected hands, or contaminated surfaces such as skin contact with an incubation period of 2 – 14 days (Zhai *et al.*, 2020) (Figure 1). The pandemic substantively affected even the most developed countries and posed a serious threat to low- and middle-income countries. It

particularly impacted nations in Africa, where people are very vulnerable to infectious diseases (Lone & Ahmad, 2020; Shastri *et al.*, 2020).

COVID-19 had caused 473,797 deaths globally as of June 24, 2020, from 9,129,146 confirmed cases. By November 25, 2022, the WHO reported 636,440,663 confirmed cases and 6,606,624 deaths worldwide (Figure 2). The highest number of cases were registered in Europe (264,065,011), followed by the Americas (181,462,467), the Western Pacific (97,712,745), Southeast Asia (60,621,308), the eastern Mediterranean (23,187,814), and Africa (9,390,554) (WHO, 2023). The first confirmed COVID-19 case in India was reported in Kerala on January 30, 2020, making India the first in Asia and the fourth globally to record confirmed cases (Chehal *et al.*, 2021). The situation escalated rapidly, with cases of infection rising from 32 on March 2, 2020, to a peak of 2,738,957 on May 3, 2021. Significant surges were also observed on specific dates with 148,249 cases on March 15, 2021, and 2,115,110 cases on January 17, 2022, followed by a critical decline. India recorded 456,183 confirmed cases during the early months of the pandemic and confirmed 14,476 deaths, primarily in Maharashtra and Delhi (Aneja & Ahuja, 2021).

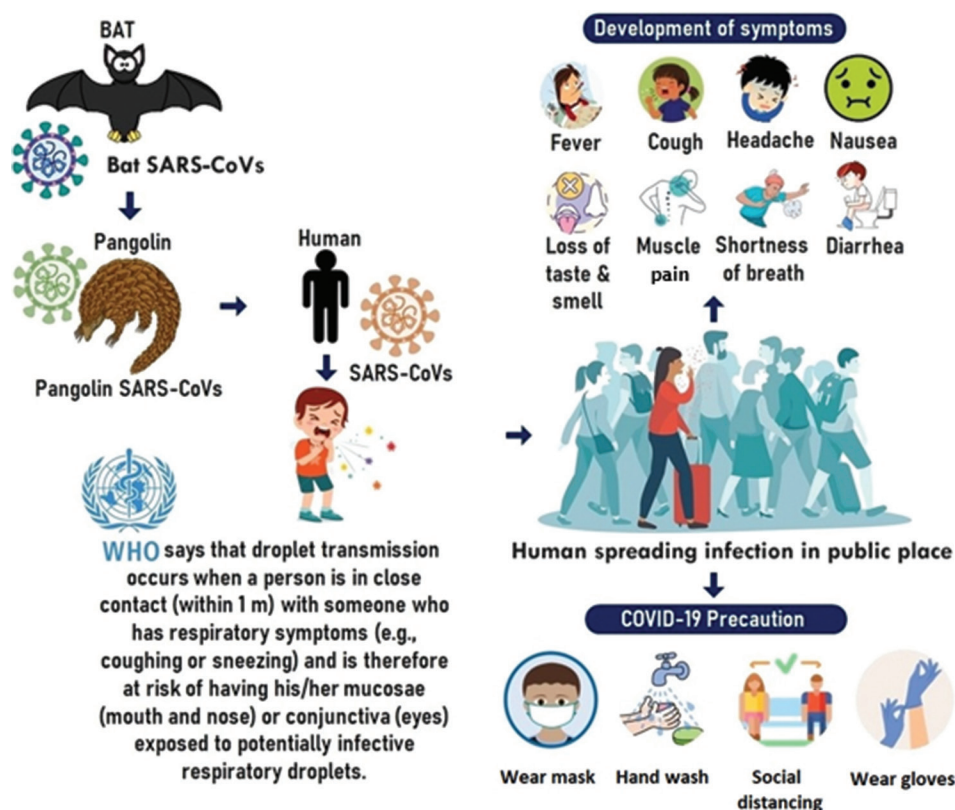


Figure 1. Infographic of COVID-19 infection. Image created by the authors
Abbreviation: SARS-CoV-2: Severe acute respiratory syndrome coronavirus 2.

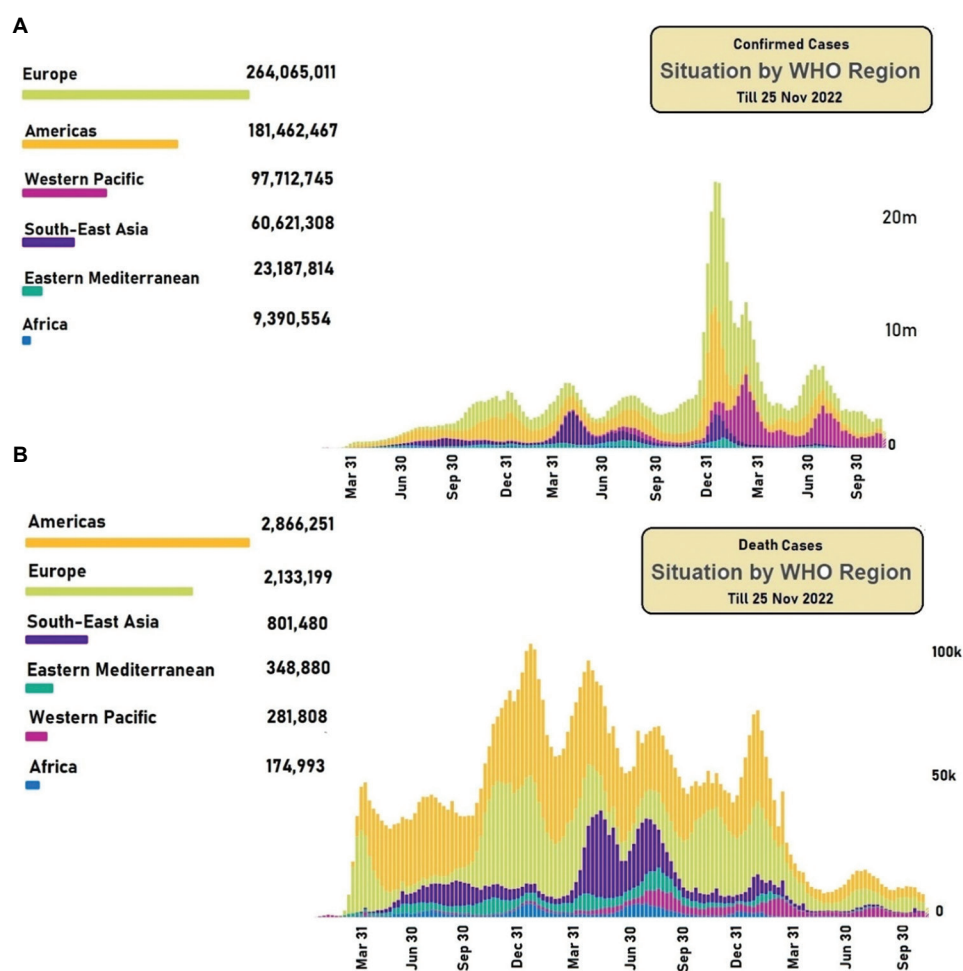


Figure 2. WHO Coronavirus (COVID-19) (November 25, 2022). (A) Presentation of infected cases in different regions of the world. (B) Graph depicting the total number of mortalities in different regions of the world
Abbreviation: WHO: World Health Organization.

The rapid global spread of COVID-19 highlighted its contagious nature, prompting nations including India to enforce travel restrictions, social distancing, and strict lockdowns. These measures were vital to curbing infections, especially during the first wave. Notably, countries with shorter lockdowns witnessed more cases per person but registered a lower average fatality rate (5.45%) than nations with longer lockdowns, which reported a higher average rate of 12.70% (Coccia, 2021). However, self-isolation and quarantine mandates caused increased mental health issues including loneliness, anxiety, depression, and harmful behaviors (Kumar & Nayar, 2021) exacerbated by continuous newscasts about the outbreak (Li *et al.*, 2020). The fear of quarantine and social stigma caused many people in India to avoid testing (Rewerska-Juško & Rejda, 2022). The impact of the pandemic was widespread, affecting healthcare workers, the economy, travel, the stock markets, factories, migrant labor, and research. The extensive and

disruptive effects of SARS-CoV-2 have marked it as one of the most significant pandemics in history (Singh & Yi, 2021). Therefore, the present review explored and assessed the impact of COVID-19 on diverse domains such as the economy, society, environment, education, and health to understand the widespread effects of the pandemic. This review intensively encompassed myriad aspects of the pandemic and would particularly benefit readers seeking a comprehensive understanding of this period of human history.

2. Material and methods

This systematic review was conducted using multiple databases including Google Scholar, PubMed, Springer, and Elsevier. It encompassed the period spanning February 2019 to December 2022. The search strategy entailed employing specific keywords and phrases related to the impact of COVID-19: for instance, “COVID-19

and health,” “COVID-19 and education,” “COVID-19 and economics,” and “COVID-19 and environment.” We restricted our evaluation to articles published in English to ensure comprehensive and unbiased coverage. Thus, we utilized language filters to exclude non-English publications.

The present review incorporated both primary and secondary sources including scientific papers, meta-analyses, systematic reviews, and peer-reviewed articles. We excluded unpublished, incomplete, or partially available data as well as publications in languages other than English to mitigate potential biases and maintain

consistency and reliability. The relevant authors (A.S., K.S., A.C.P., G.K., S.B., A.J., and S.S.) independently verified the validity of the included data to ensure accuracy.

In addition, we efforted substantially to eliminate any undesirable characteristics that could affect the study’s outcomes. The preferred reporting items for systematic reviews and meta-analyses (PRISMA) diagram (Figure 3) illustrates the flow, identification, and screening process utilized for the study and demonstrates our systematic approach to controlling variables and reducing bias in our review.

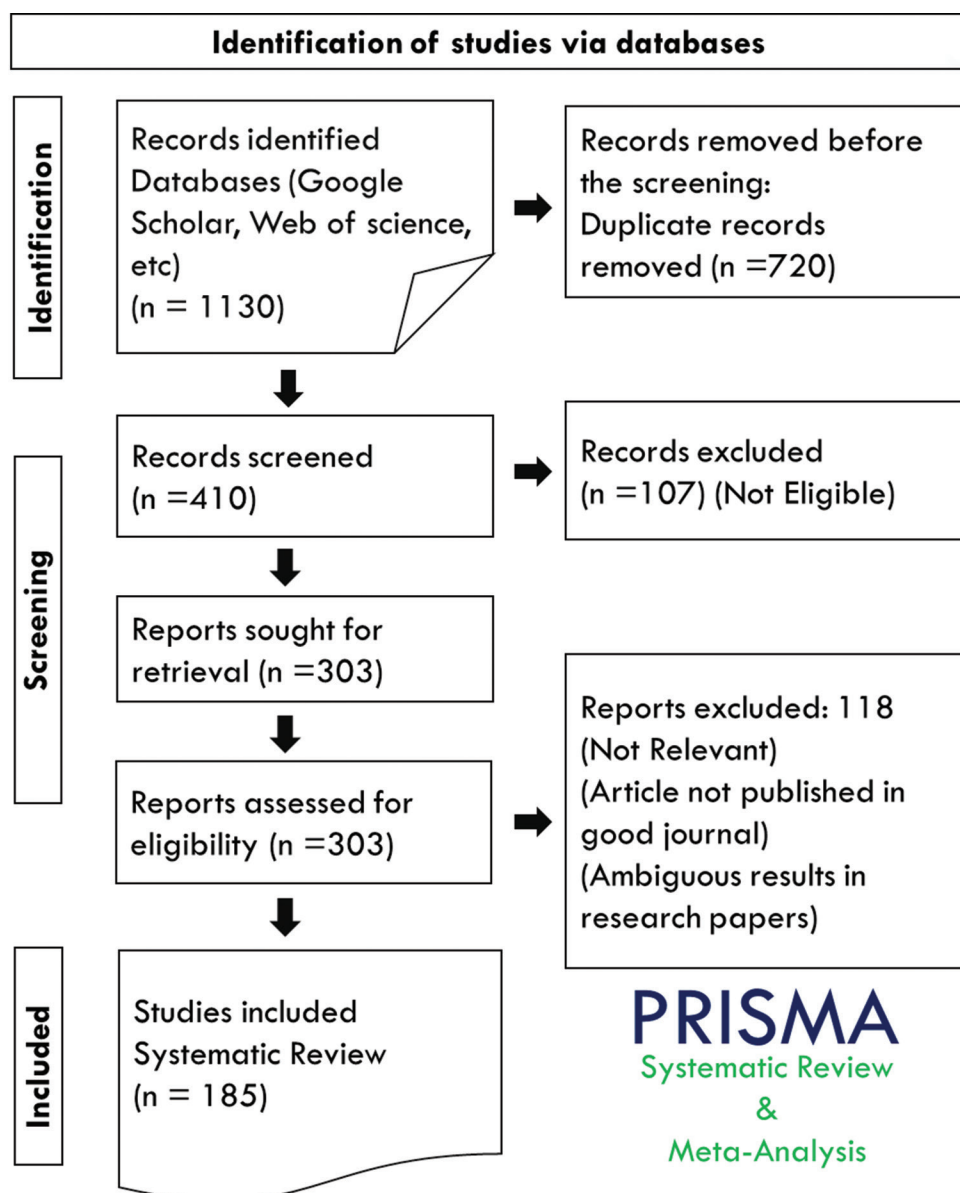


Figure 3. Preferred reporting items for systematic reviews and meta-analyses flow diagram representing the schematic representation of data extraction from online databases

3. Results

After completing our systematic literature survey, we identified and evaluated a comprehensive array of studies relevant to the impact of COVID-19. Initially, we identified 1,130 articles; however, this number was honed to 185 articles after excluding irrelevant studies. The PRISMA flow diagram (Figure 3) illustrates the stream of information through the different phases of the review including the identification, screening, and inclusion of studies.

The literature highlighted that the COVID-19 pandemic critically disrupted life worldwide, affecting nearly every aspect of human life and resources (Hiscott *et al.*, 2020). A key question consequently emerged: was this virus novel or related to earlier viruses? Coronaviruses have infected humans since the 1960s, and typically cause mild respiratory symptoms or the common cold. However, the recent SARS-CoV-2 outbreak represented the third major occurrence of respiratory illnesses triggered by this virus family in the last 20 years and significantly disrupted the global socioeconomic system.

The first known case of pneumonia caused by SARS-CoV was reported in Foshan, China, in November 2002 (Zhong *et al.*, 2003). More than 8,000 cases had been recorded by July 2003, resulting in 774 deaths. A small outbreak in 2004 added four more cases that generated symptoms such as fever, headaches, coughing, and shortness of breath (Zhao, 2007). MERS-CoV emerged in Saudi Arabia in 2012, affecting more than 2,494 people and causing 858 deaths. MERS-CoV was less contagious than SARS-CoV but was more lethal: it resulted in kidney failure in patients (Baharoon & Memish, 2019; Chakraborty & Maity, 2020). A new coronavirus, SARS-CoV-2, was identified in Wuhan, China, in December 2019. This coronavirus instigated a pandemic that resulted in millions of reported fatalities by early March 2021.

3.1. Structure and genome of COVID-19

Structurally, the COVID-19 virus is spherical and features single-stranded (positive-sense) RNA linked to a nucleoprotein in a matrix protein-encased capsid (Figure 4A). The viral genome contains distinguishing traits such as a different N-terminal portion inside the spike proteins; the genes for the key structural proteins are arranged in the 5'-3' order as S, E, M, and N5. Several minor ORFs can be identified downstream of the nucleocapsid gene and between the conserved genes (envelope, ORF1ab, membrane, spike, and nucleocapsid) (Figure 4A) (Mousavizadeh & Ghasemi, 2021). The virus's ability to adapt to a new environment/host tropism through mutations in the spike protein creates unwanted situations for researchers who try to cope immediately with

the virus (Figure 4B) (Li, 2016; Graham & Baric, 2010). The spike protein binds to acetyl-CoA dehydrogenase 2; thus, it initiates the COVID-19 infection and permits it to penetrate cells and cause the contagion. Figure 5 presents the detailed life cycle of the virus. However, an intriguing question arises: was the possibility of susceptibility to infection caused by a risk factor, or did it result merely from exposure?

3.2. Susceptibility to COVID-19 infection

Varied risk factors influence one's susceptibility to COVID-19 and can be broadly categorized as genetic and environmental features (Singh *et al.*, 2022). Environmental factors and lifestyle choices function significantly in modifying vulnerabilities determined by genetic pre-dispositions. For instance, severe or pre-existing conditions such as renal disease or anemia are linked to a higher risk of severe COVID-19 outcomes (Tao *et al.*, 2021). Hospitalization-related factors such as the need for supplemental oxygen and severe vitamin D deficiency are associated with an increased risk of mortality (Bhargava *et al.*, 2020; Carpagnano *et al.*, 2021). Nutritional deficiencies have been observed in a substantial proportion of COVID-19 patients, particularly zinc insufficiency. Such deficits generate greater complications, increase the risk of death, and mandate prolonged hospital stays (Jothimani *et al.*, 2020). In addition, hormonal changes have been noted in males who have recovered from the infection: Low levels of luteinizing hormone, 17 β -estradiol/estrogen, and testosterone indicative of hypogonadism were observed at a 7-month follow-up (Salonia *et al.*, 2022).

COVID-19 impacts more than just the lungs; it affects several vital organs and can induce varied comorbid conditions. For instance, Khoo *et al.*'s (2021) observational study found that COVID-19 patients exhibited lower levels of thyroid-stimulating hormone and free thyroxine (FT4) compared to uninfected individuals. Diabetes is another major concern: Diabetic patients with COVID-19 were at higher risk of intensive care unit admission but no significant increase was noted in their mortality rates (Al-Salameh *et al.*, 2021). Metz *et al.* reported a higher risk of cesarean delivery, premature births, and hypertensive disorders in pregnant patients with severe COVID-19 (Metz *et al.*, 2021). Neurological issues were also observed, for instance, stroke, neuromuscular problems, altered mental status, anosmia, headaches, seizures, encephalopathy, and encephalitis. Some patients also experienced mild to moderate confusion (García-Azorín *et al.*, 2021; Abildúa *et al.*, 2021).

Mortality rates from COVID-19 were notably higher for individuals with specific health conditions: for instance,

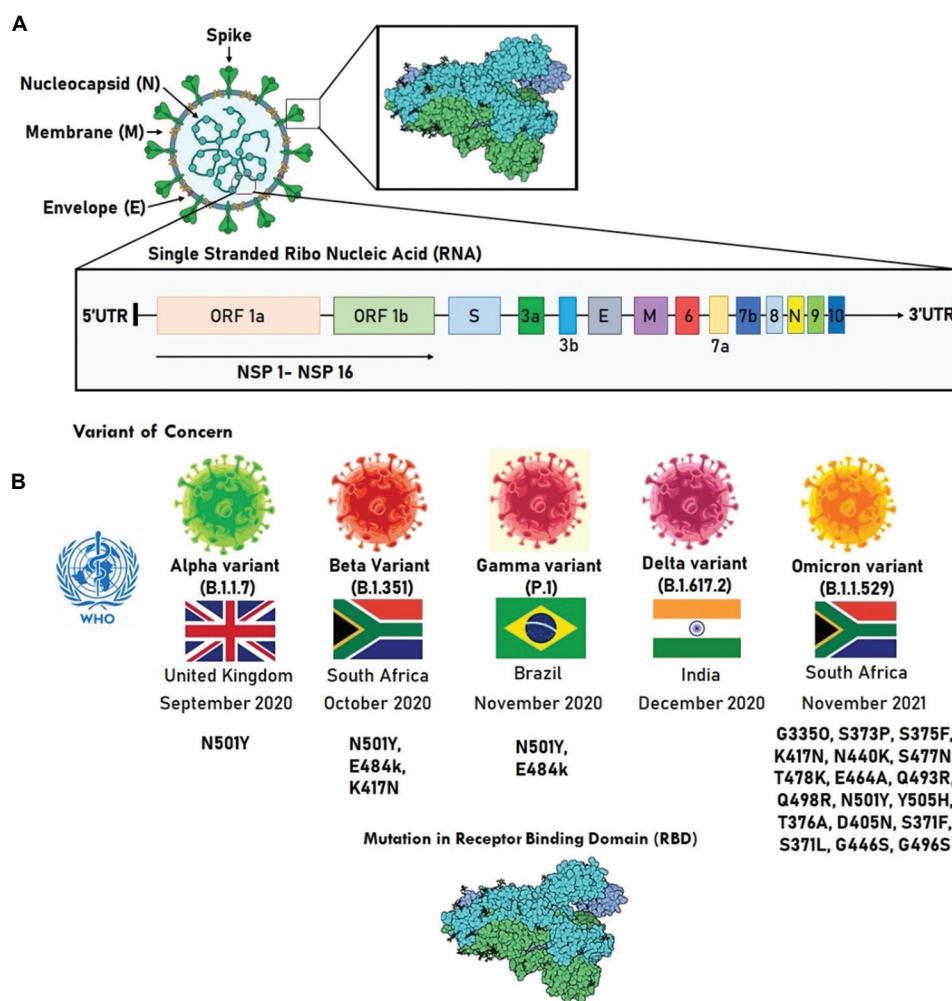


Figure 4. Different COVID-19 variants from different countries. (A) Structure of COVID-19, including the component of genes of COVID-19; (B) Variant of concern. Image created by the authors

people who were older, male, or obese (Docherty *et al.*, 2020), or who suffered from pulmonary hypertension, chronic cardiac conditions (Pagnesi *et al.*, 2020), diabetes, chronic obstructive pulmonary disease, liver diseases, and chronic kidney disease (Iaccarino *et al.*, 2020). Patients at lower risk of death also often experienced persistent symptoms, and up to 70% of such individuals exhibited organ damage 4 months after their initial symptoms. The affected organs included the heart, lungs, kidneys, liver, pancreas, and spleen (Dennis *et al.*, 2021). Vaccines significantly reduced mortality rates but debates persist about whether COVID-19 can be eradicated or could reemerge (Murray, 2022).

4. Impact of COVID-19

Governments worldwide implemented strict lockdowns when the pandemic began, and self-isolation was mandated

to curb the spread of COVID-19. These measures were effective in preventing infections (Coccia, 2021) but they also exerted a significant negative impact. As Figure 6 illustrates, the lockdowns affected many sectors including business (Donthu & Gustafsson, 2020), economy, health, and education (Jiang *et al.*, 2021; Shastri *et al.*, 2020; Nicola *et al.*, 2020; Haleem *et al.*, 2020). The sections that follow detail how strict lockdowns adversely impacted diverse aspects of society and affected resources.

4.1. Mental and emotional health

Mental health is defined as the state of well-being in which individuals can manage daily challenges and contribute constructively to society (World Health Organization, 2004; Galderisi *et al.*, 2015). The COVID-19 pandemic disrupted this balance, primarily because preventive measures such as lockdowns confined people to their

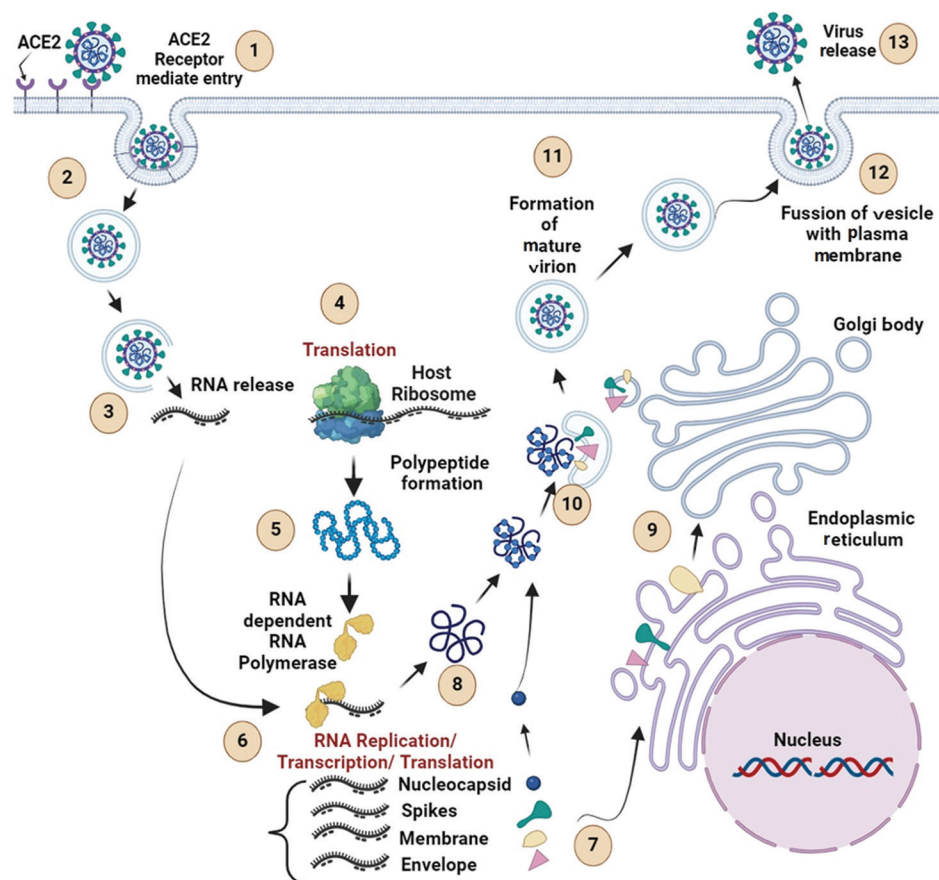


Figure 5. Schematic representation of the life cycle of COVID-19: (1) Membrane fusion at the plasma membrane following endocytosis; (2) This fusion is mediated by conformational changes in the glycoprotein induced by angiotensin-converting enzyme 2 (ACE2) binding; (3) After entering the host cell, COVID-19 releases its genomic ssRNA into the cytoplasm; (4) Here, the genome RNA is first translated into viral replicase polyprotein; (5) These polyproteins are later on are further cleaved by viral proteases into a total of 16 non-structural proteins (NSP) that form a replication-transcription complex (RTC); (6) The negative-sense (- sense) genomic RNA is generated and used as a template to produce positive-sense (+ sense) by RTC; (7) S protein membrane (M) protein and envelope (E) protein are transcribed, subsequently translated in the rough endoplasmic reticulum (RER), and transported to the Golgi complex; (8) The nucleocapsid (N) structural protein and viral RNA are replicated, transcribed, and synthesized in the cytoplasm; (9) In the RER and Golgi complex, the COVID-19 glycoprotein is subjected to cotranslational and post-translational processing including signal peptide removal, trimerization, extensive glycosylation and subunit cleavage; (10) The N protein is subsequently associated with the positive-sense genomic RNA to become a nucleoprotein complex (nucleocapsid), which is further assembled together with S, M, and E proteins as well as other viral proteins and followed by budding into the lumen of the ER-Golgi intermediate compartment to form mature virions; (11 – 13) The mature virions then reach the plasma membrane where they become fused with plasma membranes and exit the cell through exocytosis. Image created by the authors.

homes except to accomplish essential tasks such as visiting medical facilities and buying provisions (Park *et al.*, 2022). Such restrictions increase stress, confusion, and anxiety in individuals (Brooks *et al.*, 2020; Lima *et al.*, 2020). Notably, these issues were often more severely experienced by people without pre-existing mental health conditions than people who already suffered from mental health problems (Kang *et al.*, 2020; Qiu *et al.*, 2020; Singh *et al.*, 2023).

Studies linked increased social media use during lockdowns to reduced physical activity, which was associated with health problems such as early aging,

obesity, cardiovascular issues, muscle atrophy, bone loss, and decreased aerobic capacity (Kang *et al.*, 2020; Qiu *et al.*, 2020; Maugeri *et al.*, 2020). Financial difficulties, job losses, and separation from family members because of the mandated restraints also heightened depression and anxiety in people (Jean-Baptiste *et al.*, 2020). The impact was particularly profoundly sensed by children and adolescents who sensed increased anxiety and uncertainty because of school closures that disrupted their daily routines, social interactions, and access to educational resources (Imran *et al.*, 2020; Guan *et al.*, 2020).

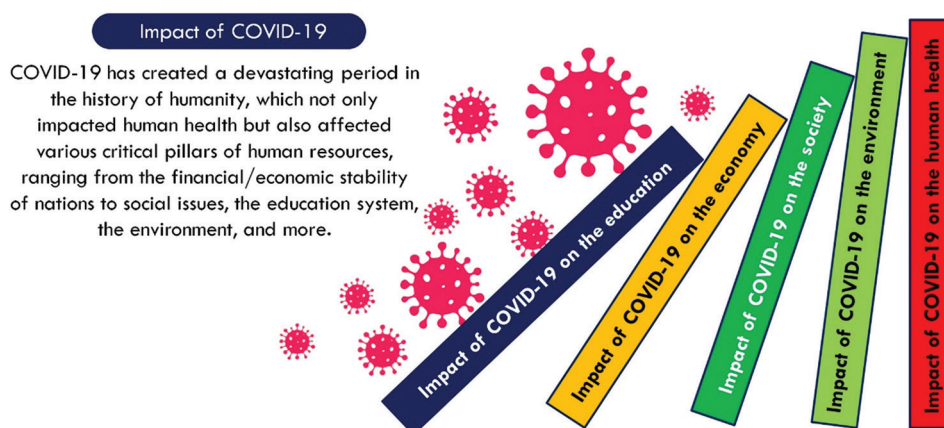


Figure 6. Graphical representation of the impact of COVID-19 on different aspects of human societies. Image created by the authors
Abbreviation: COVID-19: Coronavirus disease 2019.

4.1.1. Social media and psychological well-being

In the last decade, social media has become a crucial tool for raising public health awareness; this role intensified during the COVID-19 pandemic as people increasingly turned to platforms such as Facebook, Twitter, YouTube, Instagram, Snapchat, and WhatsApp for information. Such usage surged 87% after lockdowns were announced and helped to reduce mental stress, supported online learning, and facilitated remote work (Chand, 2021). However, the negative aspects of social media became more apparent as the pandemic progressed. People began to share misinformation disseminated by unreliable sources and believed false assertions about the prevention and treatment of COVID-19 despite the availability of accurate guidance from official health websites. Unofficial and religious pages propagated misrepresentations that became linked to poorer psychological well-being (Hernández-García & Giménez-Júlvez, 2020).

The adverse effects of social media were particularly severe in developing countries such as India. Panic fueled on social media platforms caused shortages of essential items such as masks and hand sanitizers even before the first COVID-19 case was reported. False claims were made about the virus's survival on surfaces and its airborne transmission and exaggerated news were distributed about mass casualties in China. Such fabrications generated widespread fear and anxiety. Moreover, stress and discrimination intensified because of rumors about extended lockdowns and racism toward people from India's northeastern regions who were mistakenly associated with Chinese origins (Kadam & Atre, 2020). Further, the elderly were significantly psychologically and physically distressed. The rapid spread of unscientific information altered societal perceptions and resulted in

the increased emotional abuse of older citizens, causing heightened depression in this population (Anwar *et al.*, 2020). Thus, social media facilitated communication during the pandemic but also enabled the dissemination of false information, exacerbating mental and emotional health difficulties.

4.1.2. COVID-19 and the mental health of teachers

The pandemic significantly impacted the educational sector worldwide, prompting a swift transition from offline face-to-face learning to online remote education. This shift presented both teachers and students with several challenges because the rapid adoption of information and communication technologies became central to teaching and learning processes. The mental health of teachers, who are often regarded as “nation builders,” became a major concern. Poor mental health is often associated with elevated stress levels and can undermine the effectiveness of teaching-learning processes (Borrelli *et al.*, 2014). Consequently, changes in the working conditions of teachers could be correlated with variations in their mental health states (Fernández Cruz *et al.*, 2018).

The Internet provided valuable educational tools; however, inadequate technical abilities and unreliable digital infrastructures intensified the learning disparities that existed across regions (Hidalgo-Andrade *et al.*, 2021). Many educators were unprepared for the technological challenges posed by the pandemic. Teachers struggled with online education because they were insufficiently prepared and lacked the requisite technological capabilities and pedagogies for remote teaching. Stress levels increased as teachers faced the prospect of disengaged students who were unresponsive to assignments and distracted by their surroundings (Hassan *et al.*, 2020). The psychological well-being of teachers was also affected by personal issues

such as concerns about their future and family health, the pressure to satisfy the expectations of students, parents, and school administrations, financial difficulties, and the need to balance professional and personal or family responsibilities (Jakubowski & Sitko-Dominik, 2021). These factors collectively contributed to the declining mental and emotional well-being of teachers.

4.1.3. COVID-19 and the mental health of children and adolescents

Schools function crucially in the lives of students, providing them with educational resources as well as valuable opportunities to interact with teachers and peers. However, children often become less physically active on weekends or during summer breaks when they are not in school. They tend to spend more time facing screens, experience irregular sleep patterns, and consume less nutritious food. These changes can contribute to weight gain and poor cardiorespiratory health (Brazendale *et al.*, 2017). The COVID-19 pandemic intensified such problems by imposing physical limitations and social isolation on children and adolescents. Thus, it significantly impacted every aspect of the lives of youngsters. Relatively fewer children contracted the disease but the more wide-ranging effects of social exclusion, school closures, and isolation exerted profound negative effects on their emotional and mental well-being (Meherali *et al.*, 2021).

The shift from face-to-face learning to online education introduced substantial stressors for children, particularly in developing nations such as India in which online teaching was not widely established before the pandemic. According to UNESCO, school closures affected 320 million students in India and nearly 1.725 billion students globally by May 18, 2020. Key obstacles included the paucity of digital devices, insufficient digital skills, and unreliable Internet connections (Hassan *et al.*, 2020). Children and adolescents in Jammu and Kashmir experienced severe Internet connectivity issues in the early stages of COVID-19 because of restricted 4G access, which further hampered their education. As a result, the children faced increased pandemic-related stressors such as limited educational resources, fear of infection, boredom, inadequate personal space at home, and family financial losses. These factors contribute to heightened anxiety, depression, sleep disturbances, appetite changes, and social dysfunction in children and adolescents (Brooks *et al.*, 2020).

4.1.4. COVID-19 and the mental health of medical professionals

Health-care professionals (HCPs) faced an elevated risk of burnout during the pandemic and were significantly more likely to experience stress, anxiety, and depression.

The surge in their caseloads necessitated longer working hours, often with limited resources and questionable infrastructure (Shigemura *et al.*, 2020). Personal protective equipment (PPE) was crucial for their safety but it also caused great physical discomfort and sometimes resulted in breathing difficulties. Many HCPs felt unprepared for their roles because of the absence of pre-existing guidelines or evidence-based treatments for the new virus (Huang *et al.*, 2020). Medical care professionals (MCPs) also feared infecting loved ones and concerns about autoinoculation further compounded their stress (Kang *et al.*, 2020). Many HCPs isolated themselves from their usual social circles and changed their daily routines to safeguard others; in some cases, they even separated themselves from their families. Notwithstanding these drawbacks, the pandemic elevated HCPs to superhero status, enhancing the perceived value of their roles and boosting their job satisfaction. However, public recognition also increased pressure on them and reduced room for error. The media portrayal of HCPs as heroes was driven by the sensational nature of the pandemic and underscored the need for positive reinforcement and acknowledgment (Sandesh *et al.*, 2020). Consequently, the COVID-19 pandemic evidently exerted a profound negative impact on the mental health of a significant number of healthcare workers.

4.1.5. COVID-19 and migrant workers

The impact of the pandemic was especially severe on migrant workers and the homeless, who were often bereft of stable shelters and were thus more vulnerable to social, psychological, and emotional trauma (Verma & Prakash, 2020). Many migrants were already among the most marginalized social groups and their survival depended on daily wages. Thus, their immediate anxieties during the pandemic included securing food, housing, and health care even as they grappled with the fear of contracting or spreading infection, worried about the loss of wages, and agonized about the safety of their families (Singh, 2020). Besides such challenges, migrants also sometimes confront negative reactions and intimidation from local communities.

In response, the state and central governments in India attempted to mitigate such issues by providing shelter and food to migrant workers. However, the effectiveness of these measures remains uncertain. Media reports frequently highlighted the struggles of migrants traveling long distances to return home on foot or by bicycle. According to the International Labor Organization, the pandemic resulted in a 6.7% reduction in working hours globally, and 195 million full-time jobs were lost in the second quarter of 2020 alone (International Labour Organization, 2020). A total of 255 million full-time jobs

and working hours were lost by the end of 2020 (Raut, 2020). Tragically, many individuals died en route or upon reaching their destinations (Kumar *et al.*, 2020). Conflicts between government agencies were also reported about the logistics of returning migrants to their places of origin. These cumulative factors caused migrant workers significant mental distress.

4.2. Impact of COVID-19 on society

Physical distancing instigated new survival patterns during the COVID-19 pandemic. People stayed home, developed new hobbies, and focused more on work than on household chores. This increased time at home strengthened family bonds because individuals who previously struggled to connect because of work commitments were now compelled to spend more time together. Mobile phones and the Internet became essential for continued interactions and these means helped people to adapt to new routines and balance their professional tasks with their household responsibilities (Verma & Prakash, 2020).

However, the initially positive experience of being together in proximity soon succumbed to mental health challenges. In particular, children suffered because they were forced to isolate from their peers, stay at home, and adhere to social distancing norms, which changed their brain architecture and induced psychological distress (Gautam *et al.*, 2020). The elderly also experienced significant mental health problems, particularly exhibiting depressive symptoms that were exacerbated by their difficulties with smartphones and Internet services and their restricted access to outpatient clinics and essential medications (Nikolakakis *et al.*, 2019; Mansour *et al.*, 2020; Yang *et al.*, 2020). Frontline workers including police officers and healthcare personnel experienced heightened stress, anxiety, and depression because of the demands imposed on them by the pandemic (Chen *et al.*, 2020). Women reported significantly higher levels of stress, anxiety, and post-traumatic stress disorder symptoms, and displayed more severe overall psychological effects (Wang *et al.*, 2020a & b; Liu *et al.*, 2020a & b; Daulay, 2021). In addition, mothers participating in homeschooling for children with autism spectrum disorders faced challenges such as coping with maladaptive behaviors, inadequate teaching skills, and insufficient support, which led to increased parental distress.

4.3. Impact of COVID-19 on education

The first wave of the COVID-19 pandemic presented universities and colleges worldwide with the challenge of balancing infection control against their educational responsibilities. The debate about whether to continue in-person classes or switch to online instruction became

focal and many institutions speedily adopted remote learning technologies as a viable alternative to physical presence in class. Universities and private sector educational institutions swiftly devised effective strategies to transition to virtual learning systems (VLS) despite numerous complexities, limited resources, and technology constraints. The employed strategies included professional training for faculty, staff, and students along with the institution of quality assurance measures to ensure the effectiveness of online education (Maqsood *et al.*, 2021).

Recent research endeavors have highlighted diverse approaches to delivering science, technology, engineering, math, and medicine education via teaching laboratories in virtual or remote environments. Bhute *et al.* (2021) suggested that the amalgamation of technologies such as IoT sensors and augmented or virtual reality can revitalize traditional labs, enhance student experiences, and inspire innovative pedagogies. In the context of Indian business schools, Chaturvedi *et al.* referenced concepts mooted by John Dewey, Kolb, Jack Mezirow, and Jean Piaget and identified six key teaching techniques that could help construct successful integrated learning environments. Their suggestions included (1) utilizing online knowledge repositories and virtual labs to transform digital spaces, (2) shaping educational practices through critical thinking, (3) offering diverse course genres on demand, (4) enhancing resilience through active participation and conflict resolution, (5) fostering focused and inquisitive education, and (6) embracing experiential education that values classroom diversity (Chaturvedi *et al.*, 2021). Such initiatives intend to encourage academics and educators to develop effective pedagogies tailored to the needs of Indian student populations.

Medical education was substantively disrupted during the COVID-19 pandemic because the in-person learning and practical experience crucial for specializations such as dentistry (Chang *et al.*, 2021), plastic surgery (Cho & Hong, 2021), and clinical radiography (Ofori-Manteaw *et al.*, 2022; Chen *et al.*, 2022) were severely impacted. The pandemic jeopardized the opportunities students could avail of to acquire essential hands-on knowledge and precluded the inculcation of skills typically acquired through active, face-to-face activities. Co *et al.* recommended the development of new tools and software to address such challenges: for instance, e-learning portals, online anatomy education, surgical skills training, and virtual reality platforms that could facilitate learning through online lectures (Co *et al.*, 2022). Clinical instructors adapted to the absence of real patients by themselves simulating patients and utilizing clinical case scenarios incorporating genuine patient imagery and electronic medical records (Park *et al.*, 2021).

Tuma *et al.* emphasized the need to replace traditional face-to-face activities with alternative surgical curricular components (Tuma *et al.*, 2021). Ng (2021) explored pre-registration adjustments in the Medical Radiation Science curriculum in response to the pandemic, noting their impact on varied stakeholders. Smith & Farra (2022) highlighted significant healthcare reforms triggered by the pandemic, including changes in nursing practices and education. Investigations comparing traditional classroom settings with VLS indicated that online learning can prove more effective because it offers benefits such as easier tracking of student progress, increased engagement, scheduling flexibility, enhanced efficiency, and improved academic collaborations (Hong *et al.*, 2021).

VLS emerged as crucial tools that allowed educational continuity during the pandemic but their implementation posed several challenges. Principal problems included technological difficulties, high costs, low student engagement, and trouble with focus maintenance. Specific concerns for medical students included two-dimensional representations in anatomy or surgical skill courses, limited time to engage in academic discussions, and the refusal of health institutions to accept trainees for practical experience stints during the pandemic (Ofori-Manteaw *et al.*, 2022). VLS cannot completely replace traditional in-person instruction but they offer notable advantages and serve as valuable educational aids in difficult times (Onigbinde *et al.*, 2021; Hilburg *et al.*, 2020). Hao *et al.* underscored the growing significance of Internet-based learning for medical and nursing students, both now and in the future (Hao *et al.*, 2022). Discrete research groups (Patil *et al.*, 2022; Chen *et al.*, 2022) have also highlighted the rapid advancements and improvements in radiology education during the pandemic. A hybrid approach combining simulation-based innovations, virtual readouts, virtual teaching seminars, and remote work could potentially refine the traditional model of in-person instruction by delivering flexibility, increasing convenience, and offering residents more wide-ranging exposure (Patil *et al.*, 2022).

The imperative of remote learning disproportionately impacted children belonging to racial minorities, living in poverty, requiring special education, or speaking English as a second language (Hoofman & Secord, 2021; Garg *et al.*, 2020). Studies conducted across different regions have revealed that inadequate online education infrastructure and the shortage of qualified personnel were significant issues. For example, the absence of sustainable online systems and limited Internet access in remote and rural areas were major obstacles in Nepal (Pal *et al.*, 2021). Similarly, Oyedotun identified critical online education

challenges in Guyana: the paucity of computers, laptops, or tablets; insufficient prior preparation of both students and educators; inadequate national infrastructure; slow Internet speeds; inconsistent electricity supply; ineffective course delivery; and poor student-teacher engagement. Issues such as long working hours, compromised deadlines, limited review capabilities, and concerns over mental health, cybersecurity, and viral exposure further exacerbated the difficulties faced by students and educators (Oyedotun, 2020).

Discrete surveys have been conducted with students in different countries: for instance, in the Kitwe district in Zambia (Mukuka *et al.*, 2021), at Western Michigan University in the United States (Al-Mawee *et al.*, 2021), and at an Afghan university (Noori, 2021). Most students articulated both good and bad aspects of distance learning, such as the loss of social connection or the flexibility of time and place (Al-Mawee *et al.*, 2021) in places where most residents did not enjoy adequate access to ICT, power, or online services (Mukuka *et al.*, 2021). Reuge *et al.* (2021) published a paper titled “Education response to COVID-19 pandemic, a special issue proposed by UNICEF: Editorial review.” This review article included comments from UNICEF on the effects of school closures, the efficacy of remote learning solutions, equality-related implications, and the prevention of learning losses, and presented ideas about more effectively reopening schools. The transition from in-person to online apparently exerted a negative impact on the well-being of students enrolled in higher education institutions in discrete nations such as Turkey (Topuz *et al.*, 2021), Pakistan (Maqsood *et al.*, 2021), and Ethiopia (Mengistie, 2021). Students in these countries suffered mostly from anxiety and depression, and such psychiatric abnormalities were more prevalent in adolescent populations. Most students with mental health difficulties belonged to migrant contexts, were female, or did not own substantial social support resources (Van de Velde *et al.*, 2021; Fretheim *et al.*, 2021).

4.4. Impact of COVID-19 on business

The COVID-19 crisis opened doors to innovation and offered many commercial organizations the opportunity to adopt new business models that could help them navigate the difficulties posed by the pandemic. However, the crisis also introduced significant obstacles for enterprises. The initial economic impact was characterized by abrupt decreases in the aggregate demand as well as supply; thus, many businesses were jeopardized and their revenues were significantly diminished (Seetharaman, 2020). For instance, Internet-based businesses found it difficult to sell their products online because lockdowns severely restricted import and export activities (Hasanat *et al.*,

2020). However, certain sectors such as the pharmaceutical, healthcare, vitamin, and herbal industries experienced growth and success during this period. Nonetheless, small businesses were particularly vulnerable: Many of them confronted the higher risk of closure as sales plummeted across nearly every industry (Apedo-Amah *et al.*, 2020). Small- and medium-sized enterprises (SMEs) were profoundly affected because the COVID-19 pandemic stagnated their growth and disrupted their supply chains, placing both businesses and individuals under immense pressure. Many small businesses were forced to shut down permanently because extended periods of inactivity caused significant financial losses (Guo *et al.*, 2020). Varied stakeholders such as regulators and public administrations executed financial support measures to address such challenges and assist SMEs. Governments worldwide launched policies to mitigate the adverse impact on SMEs. For example, the United States government passed the CARES Act, establishing several temporary programs to address the adverse economic effects of the COVID-19 outbreak (Song *et al.*, 2020).

The informal sector encompasses around 90% of India's workforce, and street vendors represent 14% of this informal labor force. The lockdown resulted in government regulations that restricted the operating hours of vendors, which severely affected their incomes. To cope with this difficulty, vendors were compelled to travel from one colony to another to reach more customers. Their circumstances worsened as many daily wage earners took to selling vegetables, which increased the number of sellers while the demand remained the same. The heightened competition forced vendors to lower prices, which further reduced their profits. Furthermore, many vendors lost regular customers who preferred to buy their provisions from nearby sellers (Meher, 2021). Similarly, the lockdown significantly impacted grocery stores in India. Panic buying triggered by news of the lockdown caused stock shortages (Jain, 2020), and many grocery stores were partially or fully closed. Thus, customers encountered long queues, unreliable deliveries, inflated prices, and limited product selections (Bandyopadhyaya & Bandyopadhyaya, 2021).

The hospitality sector is a major source of global employment and denotes a significant source of income; this statement applies particularly to the restaurant sector. The pandemic impaired human movement; thus, most eateries were closed for sit-in meals as governments increasingly enacted laws stipulating social isolation and lockdowns. Family-friendly eateries, bars, and taverns were adversely affected by such mandates. Fast food restaurants suffered the same effects and most such operations ran at <20% capacity, resulting in significant

financial losses and curtailing both direct and indirect jobs (Nhamo *et al.*, 2020).

A COVID-19-caused decrease was noted in the number of active firms in all industries including construction, personal services, hotels, restaurants, and transportation. The travel and tourism industries were among the sectors that were most severely impacted because many nations banned travel to curb the spread of infection. The exports of skin and leather products, clothing, automobiles, and footwear also decreased by more than 20% (Kalogiannidis, 2020). Transportation sectors including air, rail, road, and water transportation were all affected. The intricate supply and demand trends also significantly impacted passenger and freight transportation. Commercial flight activity diminished by almost 75% from 2019 levels by mid-April 2020, and global road transport activity reduced by nearly 50% from 2019. In addition, the UK's restrictions on public transportation decreased the number of subway trips in London by 95%. The global cessation of aviation activity made mobility extremely problematic (Abu-Rayash & Dincer, 2020).

Health-care services aimed at satisfying the health-related needs of people represented one of the fastest-growing industries in both developed and developing nations. Any health system is centered on healthcare workers, whose activities aim to improve the health outcomes of communities (George *et al.*, 2014). Healthcare workers were in high demand because of the additional work required during the pandemic. Thus, health-care systems sought and aimed to retain skilled workers to avoid staff shortages that could jeopardize the future safety of societies (Winter *et al.*, 2020) and job opportunities increased in health-care systems across the globe. Pharmacy services served as the principal foundation of public health during the recent pandemic. Essentially, the market was taken over by the Internet, and customers frequently purchased medications online. Online pharmacies offered advantages over traditional chain pharmacies but also posed some risks to patient safety. Most people prefer to procure their medicines online because of the COVID-19 lockdowns (Singh *et al.*, 2020). In sum, the health sector flourished in terms of employment and revenue during COVID-19.

The steady growth of Internet users worldwide and the growing awareness of populaces became strong drivers of the rapid expansion of online shopping before the COVID-19 outbreak. In addition, the increasing number of products launched online and the low-price offers for bulk purchases boosted online consumer purchases. Moreover, the wider variety of product options helped the expansion of global electronic markets (Boore *et al.*, 2017). Numerous restrictions were imposed on businesses and individuals

during the COVID-19 pandemic, significantly accelerating the transition toward the digitalization of commerce (Fetzer *et al.*, 2020). Studies identified e-commerce as a dependent variable because the declines or increases in the demand for online platforms were influenced largely by the global spread of the virus and the individual behaviors resulting from the pandemic (Ahmad & Hui, 2020). In addition, these two variables; the steady growth of Internet users worldwide and growing awareness of online shopping increased general interest in web-based business markets and were considered reasons for individual shifts toward web-based businesses (Li & Li, 2020).

4.5. Impact of COVID-19 on travel and tourism

Traveling is one of the best ways of meeting new people and discovering new destinations, provided journeys are free from incidences of illnesses, crimes, and disasters. Of course, the chances of such adverse occurrences are quite slim, and generally, such risks can be minimized or eliminated with accurate knowledge. Thus, travel journals are filled with unforgettable memories. A pandemic or epidemic is one of the most terrifying situations for travel-related organizers or planners, and such situations can render disease prevention difficult or impossible. Pandemics are transported between locations by travelers; thus, travelers risk contracting COVID-19 and infecting others who come into contact with them on their journeys (Hollingsworth *et al.*, 2006).

Travel-related infections have a long history. For example, almost 66% of around 8000 people infected with the SARS virus infection between November 2002 and June 2003 lived in China, and 5% resided in Hong Kong (Mackey & Liang, 2012). However, numerous other nations were also affected worldwide. The advent of the COVID-19 infection exerted a significant negative impact on the activities of diverse industries, including tourism. Scrutiny of the effects of the spread of the virus in the early stages of infection (regardless of official quarantine restrictions) would probably yield a significant drop in “social ingestion,” which encompasses dining out, domestic travel, attending cultural events, participating in trade shows, etc. (Hasanat *et al.*, 2020). It is widely known that the tourist sector substantively boosts the economy of nations (Do Valle *et al.*, 2006), and China’s inbound and outbound tourism markets are considered among the world’s most popular. The Chinese tourism industry contributes the largest proportion of the country’s revenue from domestic trips, which account for around CN¥ 5,128 billion in annual revenue for China (<https://asianjournals.org/online/index.php/ajms/article/view/213/96>).

Thus, China’s tourism industry was extremely affected by the COVID-19 outbreak in the nation (Hoque *et al.*, 2020). The tourism bureau of the Chinese government reported that the country alone brought in about \$127.3 billion in 2019 (<https://www.bloomberg.com/news/articles/2020-02-13/coronavirus-s-effect-on-tourism-will-carry-into-2021-experts-say>), and the major detrimental effect to the industry resulted from the rate of cancellations of travel and tourism agreements between China and other Asian countries (Chen *et al.*, 2018; Hoque *et al.*, 2020).

Research conducted by the World Travel and Tourism Council (WTTC) revealed that up to 75 million jobs would immediately be placed at risk because of COVID-19. The WTTC believed that the global effects of the Coronavirus pandemic would result in the daily loss of 1 million jobs in the travel and tourism sector. According to the WTTC’s projections, as many as 75 million people worldwide faced the prospect of losing their employment and the gross domestic product (GDP) of the travel and tourism sector would suffer a loss of \$ 2.1 trillion by 2020. Thus, research findings indicated that the impact of the pandemic would be adverse, considerable, and substantial (Mao *et al.*, 2010). Chinese consumption would not represent the only factor to adversely impact the tourism economy; rather, country-specific effects would accrue as the virus spread outside of China. According to the pessimistic projections of the Organization for Economic Co-operation and Development (OECD), global GDP expansion would decrease by 1.5% in 2020, with a 3.75% drop in world trade volumes. Other forecasts also posited dramatic outcomes (McKibbin & Fernando, 2020).

4.6. COVID-19 and financial losses

A nation is built by its people, and countries develop because of their economies, which depend on their citizens. However, as previously discussed, the world’s political, religious, economic, social, and financial systems changed drastically because of the COVID-19 pandemic. Therefore, the governments of different nations announced lockdowns to limit the exposure, restricting human movement (Kishore *et al.*, 2021). However, lockdowns became a negative variable that decreased the GDP of discrete nations. Research has shown that countries with quarantine lockdowns officially confirmed that day (i.e., Austria, Belgium, Brazil, Canada, Italy, and Spain) were hit harder by the financial markets than nations with a lower debt-to-GDP ratio, despite both falling by nearly 15% after the WHO declared COVID-19 a pandemic (Iglesias, 2022). It was found that longer-term lockdowns hampered GDP growth: the average contraction of GDP (index 2010 = 100) from the second quarter of 2019 to the second quarter of 2020 in countries imposing longer lockdowns

(i.e., around 2 months) was approximately 21% compared to 13% in countries imposing shorter lockdowns (i.e., around 15 days). Therefore, containment strategies based on prolonged societal lockdowns systematically weakened the economic systems of nations (Coccia, 2021; Makridis & Hartley, 2020). The non-interventionist herd immunity approach of the United States was also shown to impose high costs (Chen *et al.* 2021).

The GDP of several of the world's most important economic regions fell, including the Americas, Europe, Northeast Asia, the Western Pacific, and the Arctic. It was initially predicted that the GDP loss resulting from untreated COVID-19 could equate to a total of US\$1.4 trillion by 2030 assuming that 60% of the population would become infected in 3 years. This figure would amount to a tax of 0.6% of the GDP in 2019 (in 2010 US dollar terms) or around 7.7% of the GDP in 2019 (Verma *et al.*, 2021). According to the findings of Martinho *et al.* (Martinho *et al.*, 2021), the convergence indicators of the OECD countries from the end of 2017 to the end of 2019 were obliterated by the pandemic, particularly in the first two quarters of 2020.

After the initial decline in the GDP of different nations, researchers attempted to ascertain risk-related variables based on the published data. Scholars determined some risk predictors of a country's susceptibility to the COVID-19 shock as poor GDP dynamics and strong sensitivity of bond yields to VIX in the period before the pandemic (Janus, 2021; Iluno *et al.* 2021), GDP per capita, population density, income inequality (Feng *et al.* 2022), and discretionary measures adopted by governments (Kaufmann, 2020). These variables were thought to instigate the overall decline in the GDPs of nations (Havrlant *et al.* 2021; Azomahou *et al.* 2021; Kaufmann, 2020; Verma *et al.* 2021; Chen *et al.* 2021). Researchers also proposed high-potential recovery plans based on oil price estimates and empirical findings. Social safety net policy, economic transformation, innovation, technology revolution, fiscal restraint, and climate-friendly stability were deemed the most likely means of mitigating the amount of damage (Azomahou *et al.* 2021; Chudik *et al.*, 2021; Khurshid & Khan, 2021; Auerbach *et al.*, 2021).

4.7. COVID-19 on the environment

The global environment has deteriorated significantly; the pollutant levels and Air Quality Indices of many regions have exceeded permissible limits. However, the measures taken to curb the spread of the coronavirus and the resulting slowdown in economic activities during the pandemic exerted a notable impact on the environment. Restricted human activities positively affected water

bodies, air quality, and forests; however, negative effects were also observed because of the presence of aerosols, nitrogen dioxide (NO₂), and ozone (O₃). The Central Pollution Control Board (CPCB) in India reported a slight improvement in the quality of river water across the country during the lockdown. Major pollution sources such as industrial activities were temporarily stopped or considerably reduced. Hence, the quality of varied indigenous water supplies was expected to improve because of the reduced discharge of contaminants (Sivakumar *et al.*, 2020). The organic pollution levels in surface waters decreased during the lockdown period and oxygen saturation levels increased. However, fecal contamination levels remained unchanged (MoS and MoEFCC), largely because of consistent human waste production, which is closely linked to population density. Several studies reported improvements in the surface waters of rivers, lakes, and groundwater (Patel *et al.*, 2020; Lokhandwala & Gautam, 2020; Yunus *et al.*, 2020; Selvam *et al.*, 2020; Bhattacharya *et al.*, 2021).

Restrictions imposed on motorboat traffic during Italy's lockdown resulted in a significant improvement in water purity: for instance, the clear waters in Venice revealed seaweed and other aquatic life (Mack, 2020). The deserted beaches in Florida offered turtles a safe haven where they could lay their eggs (CNN, 2020). Similarly, the absence of crowds in the Dubai Marina allowed rare sightings of Eagle rays near the docks (CNN, 2020). The decreased industrial effluent discharge into Venice's canals resulted in cleaner waters, attracting schools of fish and swans, and creating an unusually calm and quiet urban environment (Mack, 2020). In India, the Ganges Dolphin was spotted in Kolkata after nearly three decades. This remarkable occurrence was reported by *The Times of India* on April 25, 2020, and was attributed to the decreased pollution during the lockdown (CPCB, 26 May 2020). Thus, the lockdown in India apparently yielded the unintended consequence of significantly initiating the cleaning of the Ganges River, India's most sacred yet heavily polluted water source (Times of India, 2020). The average suspended particulate matter (PM) content of Vembanad Lake also decreased by 15.9%, reflecting better water quality than the previous year (Yunus *et al.*, 2020). New York's air pollution levels decreased by about 50% because of pandemic-related measures (Henriques, 2020). Overall, the COVID-19 lockdowns contributed to a healthier aquatic environment and enhanced biodiversity across the globe.

Researchers observed significant reductions in greenhouse gas (GHG) emissions during the initial COVID-19 lockdowns because of the widespread closure of industries, transportation, and businesses. Data

obtained from continuous ambient air quality monitoring stations in major Indian cities vis-à-vis the first and second lockdown phases from March 25, 2020, to May 3, 2020, disclosed a notable decrease in pollutants. This decline in air PM was attributed to the cessation of industrial activities, reduced traffic congestion, paused construction work, and diminished human action. Specifically, PM with a diameter of $<2.5\ \mu\text{m}$ decreased by 9 – 70%, and PM with a diameter of $<10\ \mu\text{m}$ diminished by 20 – 68%. In addition, sulfur dioxide (SO_2) levels dropped by 19 – 77%, and NO_2 levels fell by 20 – 87%. This temporary improvement in air quality highlighted the environmental benefits of reduced industrial and vehicular activities during the lockdown. Scholars believe that cars and aviation contribute predominantly to emissions and respectively account for almost 72% and 11% of the GHG emissions in the transportation sector (Henriques, 2020).

A study conducted by the University of West Georgia found that carbon dioxide emissions would be reduced by ten tons if 100 students stopped traveling to school or university. State-imposed restrictions during the COVID-19 outbreak resulted in significant reductions in GHG emissions, approximating the extent noted in Italy and China (CDP, 2020). China reported a nearly 25% drop in carbon emissions during this period (Myllyvirta, 2020). Airborne pollutants such as NO_2 and CO_2 also decreased in discrete regions (McMahon *et al.*, 2020). Wang *et al.* used the community multi-scale air quality model to assess pollutant reductions in China and found that limiting traffic and industrial activity helped but the sole use of these measures was insufficient in improving air quality in poor weather conditions (Wang *et al.*, 2020). The study emphasized the need to endeavor further to achieve substantial reductions in air pollution. Another study conducted in Barcelona, Spain (Tobias *et al.*, 2020), observed decreased air pollution during the outbreak and reported notable reductions in Black Carbon and NO_2 . However, PM with a diameter of $<10\ \mu\text{m}$ evinced only a minor decrease, and increased O_3 levels were reported because of reduced nitrogen oxide titration of O_3 .

NO_2 is a highly reactive pollutant produced from the combustion of carbon fuels, primarily from vehicular emissions, and is harmful to human health even with short-term exposure. NO_2 can cause cellular inflammation, bronchial hyperresponsiveness, and respiratory issues, leading to 4.6 million deaths annually (Muhammad *et al.*, 2020; Latza *et al.*, 2009). Long-term exposure to NO_2 levels as low as 0.1 parts per million can lead to chronic conditions such as bronchitis and emphysema and affect lung function (Boningari & Smirniotis, 2016). The reduced industrial activity and fewer on-road vehicles during the COVID-

19 lockdowns resulted in a significant decrease in NO_2 levels, and reductions of up to 30% were observed globally (NASA, 2020). O_3 exerts differing effects depending on its location at the ground level or the upper atmosphere: stratospheric O_3 protects against UV rays but ground-level O_3 can be harmful. Increased O_3 concentrations were noted during the quarantine period and were linked to decreased levels of the mix of nitrogen oxides because of varied chemical reactions (Hashim *et al.*, 2021; Martorell-Marugán *et al.*, 2021). Such decreases led to significant O_3 increases in several regions: 183% in Chandigarh, India; 525% in Baghdad, Iraq; 56.3% in Spain; 14% in California, U.S.A.; and 49.8% in Wuhan, China (Fu *et al.*, 2020).

4.7.1. Impact of COVID-19 on forests

The National Wildlife Foundation has stated that paper constitutes 60% of school and university waste. In addition, 16 trees are needed to produce one ton of paper, and recycling ten tons of paper saves about 100 barrels of crude oil (Sun *et al.*, 2008). E-learning reduces the need for paper and consequently decreases the number of trees felled across the world. It also lowers costs incurred by students by digitizing registration, administration, curricula, and study materials (Sun *et al.*, 2008). A study conducted in the Netherlands found that e-learning reduced CO_2 emissions and decreased the carbon footprints of students and staff members (Versteijlen *et al.*, 2017). Moreover, e-learning helps to prevent the detrimental environmental effects of the annual discarding of 350 million printer cartridges that will take approximately 1,000 years to decompose (Sun *et al.*, 2008).

However, some negative environmental outcomes were also noted during the COVID-19 pandemic: for instance, medical waste increased. Wuhan generated 240 metric tons of biological waste daily at the peak of the outbreak peak, 6 times its average quantity (Zuo, 2020). Plastic medical masks made from polypropylene further exacerbated environmental problems because their decomposition poses problems (CDP, 2020; Robert, 2020; <https://www.who.int/news/item/01-02-2022-tonnes-of-covid-19-health-care-waste-expose-urgent-need-to-improve-waste-management-systems>).

In addition, numerous investigations were conducted to determine the presence of SARS-CoV-2 in municipal raw wastewater in the Netherlands (Medema *et al.*, 2020), the United States (Wu *et al.*, 2020), Australia (Ahmed *et al.*, 2020), France (Wurtzer *et al.*, 2020), India (Kumar *et al.*, 2020a-c), Turkey (Kocamemi *et al.*, 2020), the United Arab Emirates (Hasan *et al.*, 2021), Bangladesh (Ahmed *et al.*, 2020, Jakariya *et al.*, 2021). The results of these investigations offer a more comprehensive image of

the outbreak: SARS-CoV-2 genetic pre-dispositions were found in untreated water infected by coronaviruses and untreated sewage from insufficient wastewater treatment plants released into aquatic bodies (Chen *et al.*, 2021). The effects of natural contributors such as seawater and desert dust over Italy, Greece, and the entire eastern Mediterranean region is one reason why the levels of PM $<10\ \mu\text{m}$ did not drop as much as NO₂ and other gaseous pollutants. Thus, researchers have indicated the need for more thorough research of the causes of the decline in PM $<10\ \mu\text{m}$ (Conte *et al.*, 2020).

4.7.2. Impact of COVID-19 on aerosols

Diverse aerosols were employed to disinfect streets, buildings, and homes to combat SARS-CoV-2. However, the frequent use of such disinfectants can eliminate beneficial organisms and disrupt ecological balance (Elsaid *et al.*, 2021). Aerosol levels in the midpeninsula region of India improved despite nationwide lockdowns, compared to quantities noted before the lockdown. A significant increase was noted in aerosol loading in 2020: Aerosol optical depth (AOD) anomalies were predominantly positive even though some regions exhibited slight negative anomalies. Notably, AOD levels declined in northern and southern India, Pakistan, and Bangladesh during the lockdowns compared to climatology, whereas positive anomalies persisted in the midpeninsula region (Pandey & Vinoy, 2021). The COVID-19 pandemic was a critical historical event because it profoundly impacted human health as well as the financial, economic, social, educational, and environmental aspects of human existence. Despite the vaccine rollouts, ongoing debates persist about the potential of future COVID-19 outbreaks (Murray *et al.*, 2022). This situation underscores the need to continuously monitor the perceptions and awareness levels of populations. A pressing need also exists for enhanced communication strategies and institutional endeavors to effectively address such issues (Genovese *et al.*, 2022).

5. Strengths and weaknesses of this review

We record several strengths and weaknesses of the current state-of-the-art review that must be considered in evaluating its quality and comprehensiveness.

The strengths of our review include our detailed examination of the structure and genome of COVID-19, which accords essential insights into the infectious mechanisms and vulnerabilities of the virus. Notably, the review encompasses the impact of COVID-19 on seven diverse domains of human existence: Mental health, society, education, business, travel and tourism, financial health, and the environment. This extensive scope enabled us to deliver a comprehensive understanding of the multifaceted

effects of the pandemic on discrete aspects of human life and global systems. This pioneering review contributes significantly to the extant literature by integrating such wide-ranging effects into a single analysis. In so doing, it offers a broad yet insightful perspective on the widespread consequences of the pandemic.

Despite its comprehensive approach, the review admits to limitations resulting from its broad scope. Our attempts to address such a wide array of domains in a single article restricted the depth of the analysis that could be accomplished for each area. This extensive coverage necessitated a balance between breadth and detailed exploration and could have resulted in gaps, oversimplifications, or a less detailed treatment of some domains. The inherent challenges of incorporating a vast span of impact in a single study underscore the difficulties of comprehensively capturing the intricate and nuanced effects of COVID-19. Thus, our review delivers a valuable overview but could benefit if more focused and in-depth studies are conducted in the future to complement and enhance the knowledge of the effects of COVID-19 on specific domains.

6. Conclusion

The COVID-19 pandemic imposed devastating effects on nearly every aspect of human life. This respiratory infection overwhelmed healthcare systems worldwide and caused significant economic disruption. It affected industries such as tourism, hospitality, and retail. It also impacted education, increased loneliness, and foregrounded gaps in global governance. The pandemic accelerated digital adoption and technology use but exerted mixed environmental effects. It reduced air pollution but increased plastic waste from PPEs. The repercussions of COVID-19 were profound: they affected health, financial stability, social interactions, and other facets. Despite widespread vaccinations, debates continue about the long-term impact of this disease and the preparedness of human societies for future global health crises.

Acknowledgments

The authors express their sincere gratitude to the Department of Zoology at Cluster University Srinagar and the Institute of Human Genetics at the University of Jammu for their invaluable support and resources during the completion of this review. Their guidance and contributions were instrumental to the successful completion of this study.

Funding

None.

Conflict of interest

The authors declare that they have no conflict of interest.

Author contributions

Conceptualization: Kuljeet Singh, Agar Chander Pushap, Amrit Sudershan

Writing – original draft: Kuljeet Singh, Agar Chander Pushap, Gurinderjeet Kaur, Shikha Bharti, Arjumand Jhon, Srishty Sudershan, Amrit Sudershan

Writing – review & editing: Feroze Ahmed Dar, Bashir Ahmad Sheikh, Mehraj Bashir, Showkat Ahmad Najar

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data

Not applicable.

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